

From the laboratory bench to the lunar surface: framing outpaces the evidence scale

laboratory scale measured

route-scale framing asserted (verbatim from source)

Kato 2024

5 g specimen



"towards lunar base construction"



JP STARDUST

Lim 2023

50 g specimen



"industrial-scale lunar construction"



ESA Terrae Novae 2030+

Linke 2022

3.2 g specimen



"... very scalable" technology



MOONRISE flight payload

Gines 2023

≈10 kg batch



"viability ... for paving applications"

Lim 2021

50 g specimen



"... spacecraft landing pads"

Zhu 2026

200 g batch



"in situ construction of lunar infrastructure"

Tsubaki 2024

1–4 g per run



"self-sufficient production ... on the Moon"

Gram-to-kilogram bench evidence is framed at mission scale; the coordinates that would bridge the two are unreported